

Obstructions to Low-Dimensional Natural Matrix Interpretations for a Mixed-Base Collatz Rewrite System

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Abstract

Yolcu, Aaronson, and Heule encoded the Collatz conjecture as termination of an eleven-rule mixed binary–ternary string-rewriting system and asked whether standard termination interpretations can orient that system. We prove a complete negative result for one natural proof language. No one-stage relative affine matrix interpretation over the nonnegative integers exists in dimensions one or two when the two rules implementing the Collatz update are required to be strict and the nine base-conversion rules are weak. In dimension two the matrix entries and affine offsets are unbounded. The proof combines exact Farkas certificates, a rigidity lemma for the six carry rules, a Perron–Frobenius argument for the irreducible case, and an exhaustive symbolic classification of the remaining zero patterns. A deterministic checker reconstructs every parameterized certificate using integer arithmetic; bounded enumeration was used only to discover the certificate families and to regression-test the symbolic proof. The result does not resolve the Collatz conjecture and does not exclude higher-dimensional, arctic, tropical, dependency-pair, or multi-stage interpretations. It instead settles the unbounded two-dimensional natural one-stage case of a broader obstruction question raised by the original encoding, and illustrates a reproducible route from finite counterexample-guided search to a universal no-go theorem.

Keywords: Collatz conjecture; string rewriting; termination; matrix interpretation; Farkas certificate; computer-assisted mathematics

1 Introduction

The Collatz map sends an integer $n > 0$ to $n/2$ when n is even and to $3n + 1$ when n is odd. Its conjectured eventual arrival at 1 remains open. Yolcu, Aaronson, and Heule (YAH) translated the conjecture into the termination of a finite string-rewriting system and applied automatic termination tools to weakened variants [1]. Their construction is especially useful because it turns a familiar number-theoretic question into a sharply specified question about proof search.

Natural matrix interpretations are a standard termination technique [2, 3]. A successful interpretation for the full YAH system would prove its termination and hence Collatz. Failure in a prescribed class has the opposite meaning: it rules out a proof language, not the conjecture. YAH explicitly raised the possibility that their full system might admit no suitable matrix or arctic interpretation. This paper settles one precise part of that question.

Our principal result is a universal no-go theorem for one-stage relative natural affine interpretations in dimensions 1 and 2. Only the two rules that perform the Collatz step are required to

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decrease strictly; the nine rules that convert between bases are allowed to be weak. Thus the result addresses the rule-removal obligation that actually matters, rather than the unnecessarily strong requirement that every rule be strict.

The dimension-two proof was found in three stages. First, exact exhaustive search over small matrices produced integer dual certificates. Second, the supports of these certificates were generalized to parameterized families. Finally, the uncovered symbolic region was classified without any bound on the entries. Computation is therefore a discovery and checking device, not the source of the universal quantifier. The mathematical proof is given below; the artifact independently expands and checks the certificate algebra.

To the best of our knowledge, no earlier publication proves the unbounded dimension-two no-go theorem stated here. The scope is deliberately narrow: dimensions at least three, other ordered semirings, dependency pairs, and sequential rule-removal strategies remain open.

Contributions.

1. A scalar no-go theorem for the relative YAH obligation.
2. A complete unbounded two-dimensional no-go theorem over $\mathbb{N}$.
3. A finite collection of parameterized Farkas certificates and a deterministic exact checker.
4. A case study in converting bounded CEGAR-style proof search into a symbolic theorem while retaining failed conjectures and scope limits.

2 The rewrite system and interpretations

Let the alphabet be $\Sigma = \{f, t, 0, 1, 2, <, >\}$. The mixed-base system $\mathcal{T}$ consists of the rules in Table 1. We write words by juxtaposition.

Table 1: The YAH mixed binary–ternary system $\mathcal{T}$.

class	name	rule
dynamic	even	$f > \rightarrow >$
dynamic	odd	$t > \rightarrow 2 >$
carry	f0	$f0 \rightarrow 0f$
carry	f1	$f1 \rightarrow 0t$
carry	f2	$f2 \rightarrow 1f$
carry	t0	$t0 \rightarrow 1t$
carry	t1	$t1 \rightarrow 2f$
carry	t2	$t2 \rightarrow 2t$
left	0	$< 0 \rightarrow < t$
left	1	$< 1 \rightarrow < ff$
left	2	$< 2 \rightarrow < ft$

YAH prove that termination of $\mathcal{T}$ is equivalent to the Collatz conjecture [1, Theorem 3.17]. Let $\mathcal{X}$ denote the nine non-dynamic rules. They terminate independently, so a rule-removal proof may orient the two dynamic rules strictly and every rule in $\mathcal{X}$ weakly.

Definition 2.1 (Natural affine matrix interpretation). Fix $d \geq 1$. Each symbol $s \in \Sigma$ is assigned

$$[s](x) = M_s x + v_s, \quad M_s \in \mathbb{N}^{d \times d}, \quad v_s \in \mathbb{N}^d.$$

The upper-left entry of every M_s is positive, the standard sufficient condition for strict monotonicity in the first component. The interpretation of a word is functional composition from left to right. We use the orders

$$x \geq_c y \iff x_i \geq y_i \text{ for all } i, \quad x >_c y \iff x_0 > y_0 \text{ and } x_i \geq y_i \text{ (} i > 0 \text{)}.$$

An interpretation is *relative* for $(\mathcal{T}, \mathcal{X})$ if the two dynamic rules are oriented by $>_c$ and the other nine by $\geq_c$.

For a word $w = s_1 \cdots s_k$ write

$$M_w = M_{s_1} \cdots M_{s_k}, \quad v_w = \sum_{j=1}^k M_{s_1} \cdots M_{s_{j-1}} v_{s_j}.$$

Orientation of $\ell \rightarrow r$ for all $x \in \mathbb{N}^d$ requires $M_\ell \geq M_r$ entrywise and the corresponding weak or strict inequality between v_ℓ and v_r .

For dimension two we abbreviate

$$F = M_f, \quad T = M_t, \quad Z = M_0, \quad O = M_1, \quad D = M_2, \quad L = M_<, \quad G = M_>,$$

and let $\ell = e_0^\top L = (a, b)$, where $a \geq 1$ and $b \geq 0$. The eleven coefficient obligations are

$$FG \geq G, \quad TG \geq DG, \tag{1}$$

$$FZ \geq ZF, \quad FO \geq ZT, \quad FD \geq OF, \tag{2}$$

$$TZ \geq OT, \quad TO \geq DF, \quad TD \geq DT, \tag{3}$$

$$LZ \geq LT, \quad LO \geq LF^2, \quad LD \geq LFT. \tag{4}$$

All matrix inequalities in this paper are entrywise unless stated otherwise.

3 Exact dual certificates

Once the matrices are fixed, the offset obligations are linear inequalities in the $7d$ nonnegative offset variables. We use the following elementary form of Farkas' lemma [4].

Lemma 3.1 (Certificate criterion). *For each rule r and component i , write the offset difference as $\Delta_r^{(i)}(v)$ and its required right-hand side as $\epsilon_r^{(i)}$, where $\epsilon = 1$ exactly for component 0 of a strict rule and 0 otherwise. If nonnegative integers $\lambda_{r,i}$ satisfy*

$$\sum_{r,i} \lambda_{r,i} \Delta_r^{(i)}(v) \leq 0 \quad \text{for every } v \in \mathbb{N}^{7d}, \quad \sum_{r,i} \lambda_{r,i} \epsilon_r^{(i)} > 0,$$

then the offset system is infeasible.

Proof. The assumed orientation inequalities imply that the displayed left side is at least the positive displayed right side. The coefficient condition and $v \geq 0$ imply that it is at most zero, a contradiction. $\square$

The artifact represents a certificate as a finite list $(r, i, \lambda_{r,i})$, reconstructs every coefficient from the original rules, and accepts only if all coefficients are nonpositive and the strict right side is positive. No floating-point result is trusted.

4 The scalar obstruction

Theorem 4.1. *There is no one-dimensional natural affine interpretation in which the two dynamic rules of $\mathcal{T}$ are strict and the nine rules of $\mathcal{X}$ are weak.*

Proof. Write $[s](x) = a_s x + b_s$ with $a_s \geq 1$ and $b_s \geq 0$. The coefficient parts of five weak obligations give

$$a_2 \geq a_f a_t, \quad a_t \geq a_2, \quad a_f a_1 \geq a_0 a_t, \quad a_f a_2 \geq a_1 a_f, \quad a_t a_0 \geq a_1 a_t.$$

The first two force $a_f = 1$ and $a_t = a_2$. Cancelling positive factors in the last three gives $a_1 \geq a_0 a_2$, $a_2 \geq a_1$, and $a_0 \geq a_1$; hence $a_1 \geq a_1^2$ and

$$a_f = a_t = a_0 = a_1 = a_2 = 1.$$

Strictness of $t \rightarrow 2 >$ requires $b_t > b_2$, whereas weakness of $< 2 \rightarrow < ft$ gives $b_2 \geq b_f + b_t \geq b_t$. This is impossible. $\square$

5 The two-dimensional obstruction

We now prove the main result. There is no upper bound on a matrix entry or offset in this section.

5.1 The first certificate and its complement

Lemma 5.1 (First-row growth). *If (1)–(4) hold and $\ell F \geq \ell$, then the relative offset system is infeasible.*

Proof. The last inequality of (4) gives $\ell D \geq \ell FT \geq \ell T$. Take component 0 of the odd dynamic rule with weight a , component 1 with weight b , and component 0 of $< 2 \rightarrow < ft$ with weight 1. The combined coefficients of the offsets $2, f, t, >$ are respectively $0, -\ell, \ell - \ell F$, and $\ell T - \ell D$; every other coefficient is zero. They are all nonpositive, while the strict right side is $a > 0$. Apply Lemma 3.1. $\square$

Lemma 5.2 (Shape of the complement). *If the hypotheses of Lemma 5.1 hold but $\ell F \not\geq \ell$, then*

$$b > 0, \quad f_{11} = 0, \quad a f_{01} < b.$$

Proof. The first component of ℓF is at least a because $f_{00} \geq 1$. Thus failure occurs in the second component: $a f_{01} + b f_{11} < b$. This implies all three assertions. $\square$

5.2 Rigidity of carry inequalities

Lemma 5.3 (Diagonal and determinant rigidity). *In each of the six carry inequalities (2)–(3), the compared products have equal diagonal entries and equal determinants. Consequently, if both off-diagonal entries of the smaller product are positive, the entire inequality is an equality.*

Proof. For $FZ \geq ZF$ and $TD \geq DT$, equality of traces forces equality of the two diagonal entries. For the other four rules, cyclic invariance of trace gives the closed chain

$$\text{tr}(FO) \geq \text{tr}(ZT) = \text{tr}(TZ) \geq \text{tr}(OT) = \text{tr}(TO) \geq \text{tr}(DF) = \text{tr}(FD) \geq \text{tr}(OF) = \text{tr}(FO).$$

Hence all diagonal inequalities are equalities. If nonnegative 2×2 matrices $A \geq B$ have the same diagonal, then $\det A \leq \det B$. Applying this observation around the analogous closed chain of determinants yields equality of all determinants. For off-diagonal pairs $(x, y) \geq (x_0, y_0)$, equality of determinants gives $xy = x_0 y_0$; if $x_0 y_0 > 0$, both inequalities must be equalities. $\square$

There is a stronger form in the irreducible branch.

Lemma 5.4 (Perron rigidity). *If $F + T$ is irreducible, then all six carry inequalities are matrix equalities. In particular,*

$$FD = OF, \quad TO = DF, \quad TD = DT, \quad TFD = DF^2. \quad (\text{K})$$

Proof. Let $S = Z + O + D$ and let $\Delta_i \geq 0$ be the six carry slacks. Direct summation gives

$$\sum_i \Delta_i = (F + T)S - S(F + T) \geq 0.$$

For positive left and right Perron vectors $x^\top, y$ of $F + T$, the bilinear form of the right side is zero. A nonnegative matrix between strictly positive vectors can have zero bilinear form only if it is zero. Thus every $\Delta_i = 0$. The last identity follows by composition: $TFD = T(OF) = (TO)F = (DF)F$. $\square$

5.3 Excluding non-idempotent shapes

On the complement in Lemma 5.2, write

$$F = \begin{pmatrix} u & p \\ r & 0 \end{pmatrix}, \quad u \geq 1, \quad p, r \geq 0.$$

The following two propositions contain the structural core of the proof.

Proposition 5.5 (The $u \geq 2$ branches). *No matrices satisfying (1)–(4) and Lemma 5.2 exist with $u \geq 2$.*

Proof. There are four zero patterns for (p, r) .

If $p = r = 0$, Lemma 5.3, $LO \geq LF^2$, and $TG \geq DG$ give $d_{00} = o_{00} \geq u^2$. Writing the first column of G as $(c, d)^\top$ with $c > 0$, the first component of $TG \geq DG$ forces $d > 0$ and $t_{01} > d_{01}$. If $z_{11} > 0$, the lower components of $FO \geq ZT$ and $TG \geq DG$ reduce the upper-right component of $TD \geq DT$ to $ud_{01} \geq d_{00}t_{01}$, impossible because $d_{00} > u$. If $z_{11} = 0$, the carry equalities imply $u > t_{11} \geq d_{11}$, while the same component gives

$$d_{01}(u - t_{11}) \geq t_{01}(d_{00} - d_{11}).$$

Its left side is strictly below $t_{01}(u - t_{11})$ and its right side is strictly above it.

If $p > 0, r = 0$, rigidity gives $z_{10} = o_{10} = 0$ and, with $x = o_{00}$,

$$x \geq u^2, \quad ud_{00} + pd_{10} = ux, \quad t_{00}x = ud_{00}.$$

Thus $t_{00}x + pd_{10} = ux$ and $t_{00} \leq u$. The first component of $TG \geq DG$ forces the first column $(c, d)^\top$ of G to have $d > 0$ and $t_{01} > d_{01}$. If $z_{11} > 0$, lower carry components force T and D to have zero lower rows, and the upper-right component of $TD \geq DT$ becomes $ud_{01} \geq xt_{01}$, impossible since $x > u$. If $z_{11} = 0$, the upper-right component of $FZ \geq ZF$ and the diagonal carry equations first rule out $t_{00} < u$; hence $t_{00} = u, d_{10} = 0, d_{00} = x, z_{00} = x, t_{10} = 0$. The lower dynamic component gives $t_{11} \geq d_{11}$. If $t_{11} = 0$, the preceding impossible inequality returns directly. Otherwise the summed upper-right carry equations give $u > t_{11} \geq d_{11}$, whereas $TD \geq DT$ requires

$$d_{01}(u - t_{11}) \geq t_{01}(x - d_{11}).$$

The left side is strictly below $t_{01}(u - t_{11})$ and the right side is strictly above it.

The case $p = 0, r > 0$ is the transpose-shaped calculation relevant to the first column. It yields

$$t_{01} = 0, \quad t_{00} = u, \quad z_{01} = o_{01} = d_{01} = 0, \quad z_{00} = o_{00} = d_{00},$$

and then, with $A = au + br > 0$,

$$u(ao_{00} + bo_{10}) \leq Ad_{00}, \quad ao_{00} + bo_{10} \geq uA.$$

Hence $d_{00} \geq u^2$, whereas the first component of $TG \geq DG$ gives $u \geq d_{00}$.

Finally suppose $p, r > 0$. This is covered by Proposition 5.6, which in fact permits $u = 1$. The four cases exhaust all nonnegative (p, r) . $\square$

Proposition 5.6 (Fully off-diagonal branch). *There is no solution with $u, p, r \geq 1$.*

Proof. Now $F + T$ is irreducible, so Lemma 5.4 applies. Let $\lambda > 1$ be the Perron root of F ; the other root $\mu < 0$ satisfies

$$\lambda + \mu = u, \quad \lambda\mu = -pr, \quad F^2 = uF + prI. \quad (\text{L})$$

The matrix D is nonzero because $d_{00} \geq 1$.

If $\text{rank } D = 1$, write $D = cd^\top$. The commutation $TD = DT$ and identity (K), together with Perron–Frobenius for the positive matrix F^2 , imply that $d^\top F = \lambda d^\top$ and that the eigenvalue of T on the image of D is λ . The remaining carry equalities then imply $Fc = \lambda c$. For the first column g of G , put $h = Dg > 0$. The left and dynamic inequalities yield

$$\ell h = \ell Dg \geq \ell FTg \geq \ell FDg = \ell Fh = \lambda \ell h,$$

contradicting $\lambda > 1$.

If $\text{rank } D = 2$, identities (K) and the carry equalities imply that TF is similar to F^2 and FT is similar to T^2 . Traces and determinants show that T and F share the roots λ, μ . Moreover $FZ = ZF$ gives $Z = \alpha I + \beta F$ with $\alpha, \beta \geq 0$. Expressing $FTZ = ZT^2$ in an eigenbasis of F shows that the Perron eigenvector $v > 0$ of F is also a Perron eigenvector of T . Since $TD = DT$, write $Dv = \delta v$. The dynamic inequality gives $\delta \leq \lambda$, while $LD \geq LFT$ gives $\delta \ell v \geq \lambda^2 \ell v$, so $\delta \geq \lambda^2$. Again this contradicts $\lambda > 1$. $\square$

Corollary 5.7 (Exhaustive residual classification). *Every coefficient assignment not covered by Lemma 5.1 has*

$$F^2 = F, \quad F \in \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 1 & p \\ 0 & 0 \end{pmatrix} \ (p \geq 1), \begin{pmatrix} 1 & 0 \\ r & 0 \end{pmatrix} \ (r \geq 1) \right\}.$$

Proof. Lemma 5.2 gives $f_{11} = 0$ and monotonicity gives $u = f_{00} \geq 1$. Proposition 5.5 forces $u = 1$, while Proposition 5.6 forces $pr = 0$. Hence $F = \begin{pmatrix} 1 & p \\ r & 0 \end{pmatrix}$ with $pr = 0$, and direct multiplication gives $F^2 = F$. The three displayed cases are exhaustive. $\square$

5.4 Covering the three idempotent forms

The final step is offset infeasibility. Each case below is closed by one of seven parameterized certificate schemes. Their exact supports and weights are reconstructed by the artifact; Appendix A records the mathematical predicates used by the checker.

Proposition 5.8 (Lower form). *If $F = \begin{pmatrix} 1 & 0 \\ r & 0 \end{pmatrix}$ with $r \geq 1$, the relative offset system is infeasible.*

Proof. The lower-triangular calculation in Proposition 5.5, up to the step that used $u \geq 2$, remains valid at $u = 1$. It yields

$$t_{00} = 1, \quad t_{01} = 0, \quad z_{01} = o_{01} = d_{01} = 0, \quad z_{00} = o_{00} = d_{00} = 1.$$

The lower-left components of $FD \geq OF$ and $LO \geq LF^2$ force $o_{10} = r, o_{11} = 0$. The remaining carry and left inequalities force $z_{11} = t_{11} = 0$, $z_{10} = t_{10} = r$, and $d_{10} \geq r$. These are exactly the applicability conditions of certificate scheme 5; its strict right side is $5(a + br) > 0$. $\square$

Proposition 5.9 (Projection form). *If $F = \text{diag}(1, 0)$, one of certificate schemes 2–6 applies.*

Proof. Carry rigidity gives $z_{10} = o_{10} = 0$ and $d_{00} = o_{00}$. If $z_{11} \geq 1$, scheme 2 applies. Otherwise $t_{10} > 0$ would force, successively, $z_{00} = 1, z_{01} = 0$ and the impossible first-row inequality $a \geq at_{00} + bt_{10} > a$. Thus $t_{10} = d_{10} = 0$, and the remaining diagonal equations give $z_{00} = o_{00} = d_{00} = t_{00} = 1$.

Set $s = t_{01}$, $y = t_{11}$, $z = z_{01}$, $o = o_{01}$, $q = o_{11}$, $d = d_{01}$, and $e = d_{11}$. The carry and left rules imply

$$o \geq s + zy, \quad z \geq s + oy, \quad az \geq as + by, \quad qy = 0.$$

If $y > 0$, these force $y = 1, s = 0, o = z$, and $az \geq b$, which is scheme 6. If $y = 0$ and $s \leq d$, scheme 5 applies. If $y = 0$ and $s > d$, the last left inequality gives $ad + be \geq as$, hence $e \geq 1$. For $e = 1$ use scheme 3. For $e \geq 2$ the integer interval of scheme 4 is nonempty; one valid choice is

$$x = \max \{ \lceil b/2 \rceil, \lceil a(s - d)/(e + 1) \rceil \} \leq b.$$

These cases are exhaustive. $\square$

Proposition 5.10 (Upper form). *If $F = \begin{pmatrix} 1 & p \\ 0 & 0 \end{pmatrix}$ with $p \geq 1$, one of certificate schemes 4 or 12 applies.*

Proof. Lemma 5.2 gives $b > ap$. The carry rules first imply $z_{10} = o_{10} = 0$. If $t_{10} > 0$, Perron rigidity makes every carry rule an equality. Then $z_{11} = 0$, $Z = cF$ for $c = z_{00} \geq 1$, and, on writing $T = \begin{pmatrix} x & y \\ s & w \end{pmatrix}$ with $s > 0$, the carry equations reduce to

$$0 = (x - 1)(x + ps) + ys + p^2s^2,$$

which is impossible. Hence $t_{10} = d_{10} = 0$, and the diagonal equations give $z_{00} = o_{00} = d_{00} = t_{00} = 1$.

Put $s = t_{01}$, $w = t_{11}$, $z = z_{01}$, $k = z_{11}$, $o = o_{01}$, $q = o_{11}$, $d = d_{01}$, and $e = d_{11}$. The required scalar consequences are

$$\begin{aligned} z + pk &\geq p, & o + pq &\geq s + zw, & d + pe &\geq p, \\ z + sk &\geq s + ow, & o + sq &\geq p, & d + se &\geq s + dw, \\ az + bk &\geq as + bw, & ao + bq &\geq ap, & ad + be &\geq a(s + pw), & kw = qw = 0. \end{aligned} \tag{N}$$

If $w > 0$, (N) forces $k = q = 0$, then $w = 1, s = 0, z = o$. If $e \leq 1$, scheme 12 applies; if $e \geq 2$, scheme 4 applies with $x = b$. If $w = 0$, scheme 12 again applies for $e \leq 1$, while for $e \geq 2$ scheme 4 applies with $x = b$, using $ap - b < 0$. Thus every upper-form assignment is covered. $\square$

Theorem 5.11 (Unbounded two-dimensional no-go). *The system $\mathcal{T}$ has no relative natural affine matrix interpretation of dimension two in which the two dynamic rules are strict and the other nine rules are weak. Matrix entries and offsets are unrestricted elements of $\mathbb{N}$.*

Proof. If a coefficient inequality (1)–(4) fails, no interpretation exists. Otherwise, Lemma 5.1 supplies a certificate when $\ell F \geq \ell$. On its complement, Corollary 5.7 gives exactly the lower, projection, or upper idempotent form. Propositions 5.8–5.10 cover those forms by exact nonnegative Farkas certificates with positive strict right side. Lemma 3.1 excludes the offsets in every case. $\square$

Corollary 5.12. *No such interpretation exists in dimension at most two.*

6 Automation and reproducibility

The implementation has two deliberately separated layers.

Discovery. Bounded enumeration and linear programming were used to find small supports for dual certificates. Counterexamples to provisional families enlarged the search region. The numerical LP output is not used by the final verifier.

Exact checking. The Python module `sag_collatz.matrix_attack` stores the original rewrite rules, constructs word matrices and offsets, and validates all certificate rows with arbitrary-precision integer arithmetic. The only certificate with a free integer weight solves its fourteen inequalities as an exact integer interval rather than by bounded trial. A separate C++17 program exhaustively regression-tests bounded coefficient boxes.

What the checker proves. For any supplied matrix assignment, the checker validates the coefficient obligations, selects a parameterized family, and reconstructs a Farkas contradiction. It does not by itself prove that the families cover every unbounded assignment. That universal coverage is the content of Lemmas 5.2–5.4, Propositions 5.5–5.10, and Corollary 5.7. We call the result *computer-assisted*, not formally verified in Lean, Coq, or an SMT proof calculus.

The repository contains unit tests for rule expansion, exact certificate validation, adversarial mutations, the scalar theorem, bounded exhaustive counts, and structural families at coefficients far outside the discovery boxes. Reproduction commands and expected outputs are recorded in `ARTIFACT.md`.

7 Related work

Matrix interpretations for term and string rewriting were developed as a general automatic termination technique by Endrullis, Waldmann, and Zantema [2], building on automatic string-rewriting termination methods such as [3]. Relative termination and rule removal are standard ways to combine weak and strict orientation obligations [5].

YAH [1] provide the direct point of departure: they prove the equivalence between Collatz and termination of $\mathcal{T}$, automatically find interpretations for weakened systems, and identify nonexistence of suitable interpretations for the full system as a useful possibility to investigate. They also

prove a substantially broader natural-matrix obstruction for a different, previously studied *unary* Collatz rewrite encoding, even in combination with dependency pairs. That theorem does not apply to the mixed-base system $\mathcal{T}$ considered here; conversely, our theorem is limited to dimensions at most two but addresses the mixed-base encoding that makes their automated positive results possible. Our work is complementary to automatic synthesis. It certifies that one specific synthesis space—one-stage natural affine interpretations in dimensions at most two—is empty even with unbounded coefficients.

The result is not a lower bound for all matrix interpretations. Natural matrices in arbitrary dimension are already only one termination language; arctic and tropical semirings, dependency pairs, and richer interpretation schemes have different expressive power. Nor does failure in every fixed finite dimension automatically imply nontermination of the rewrite system.

8 Limitations and conclusion

Theorem 5.11 is a no-go theorem about a proof method. It neither proves nor refutes Collatz. Specifically, it leaves open:

- natural matrix interpretations of dimension at least three;
- arctic, tropical, and other ordered semirings;
- dependency-pair transformations;
- multi-stage rule-removal sequences outside the single relative obligation studied here;
- alternative rewrite encodings equivalent to Collatz.

Within its scope, however, the theorem is complete: there is no coefficient ceiling, no offset ceiling, and no unclassified zero pattern. The proof also demonstrates a useful workflow for automated reasoning. Small exhaustive search can expose recurring dual supports; exact arithmetic can turn them into trusted local certificates; and structural algebra can remove the finite search boundary. Here that process closes a concrete low-dimensional search space proposed by an influential automated approach to Collatz.

Statements and Declarations

Competing interests. The authors declare no competing interests.

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Data and code availability. All source code, exact certificates, tests, and the manuscript sources are publicly available at <https://github.com/kadirkbekov-ship-it/collatz-matrix-no-go> and in a versioned archival deposit at <https://doi.org/10.5281/zenodo.22098492>.

Author contributions. K.K. and D.K. contributed to the conception, oversight of the AI-assisted mathematical analysis, verification workflow, and review of the manuscript. Both authors reviewed and approved the submitted version.

Use of generative AI. OpenAI Codex using GPT-5.6 Sol was used as an adversarial mathematical and coding assistant for exploratory derivations, implementation, testing, and language drafting. Anthropic Claude Opus was used for a separate adversarial audit of intermediate claims and calculations. All theorem statements, proofs, computations, citations, and the final manuscript were reviewed and accepted by the human authors. Model outputs were not treated as independent evidence, and the systems are not authors.

A Certificate predicates

For auditability we summarize the seven schemes used by the universal proof. Each predicate is sufficient for the exact row emitted by the implementation; the checker expands the row against all fourteen offset variables.

Write Δ_r^i for the offset difference of rule r in component i . The seven nonnegative combinations are

$$\begin{aligned} C_1 &= a\Delta_{\text{odd}}^0 + b\Delta_{\text{odd}}^1 + \Delta_{\text{left2}}^0, \\ C_2 &= a\Delta_{\text{even}}^0 + a\Delta_{\text{odd}}^0 + b\Delta_{\text{odd}}^1 + b\Delta_{\text{f1}}^1 + \Delta_{\text{left2}}^0, \\ C_3 &= a\Delta_{\text{even}}^0 + b\Delta_{\text{even}}^1 + a\Delta_{\text{odd}}^0 + b\Delta_{\text{t2}}^1 + \Delta_{\text{left2}}^0, \\ C_4(x) &= a\Delta_{\text{even}}^0 + x\Delta_{\text{even}}^1 + a\Delta_{\text{odd}}^0 + x\Delta_{\text{odd}}^1 + x\Delta_{\text{t2}}^1 + \Delta_{\text{left2}}^0. \end{aligned}$$

For scheme 5 put $A = a + bf_{10}$; then

$$\begin{aligned} C_5 &= 2A\Delta_{\text{even}}^0 + 3A\Delta_{\text{odd}}^0 + a\Delta_{\text{f1}}^0 + b\Delta_{\text{f1}}^1 \\ &\quad + (2a + bf_{10})\Delta_{\text{f2}}^0 + b\Delta_{\text{f2}}^1 + b\Delta_{\text{t2}}^1 + \Delta_{\text{left0}}^0 + \Delta_{\text{left1}}^0 + \Delta_{\text{left2}}^0. \end{aligned}$$

Scheme 6 is

$$\begin{aligned} C_6 &= 3a\Delta_{\text{even}}^0 + 3a\Delta_{\text{odd}}^0 + a\Delta_{\text{f1}}^0 + a\Delta_{\text{f2}}^0 + 3b\Delta_{\text{f2}}^1 \\ &\quad + a\Delta_{\text{t0}}^0 + 2b\Delta_{\text{t1}}^1 + \Delta_{\text{left1}}^0 + 2\Delta_{\text{left2}}^0. \end{aligned}$$

For scheme 12 put

$$K = az_{01} + bz_{11}, \quad q = a(o_{01} - f_{01}(1 - o_{11})), \quad H = af_{01} + d_{11}K + b.$$

Then

$$\begin{aligned} C_{12} &= q\Delta_{\text{even}}^1 + 3a\Delta_{\text{odd}}^0 + H\Delta_{\text{odd}}^1 + a\Delta_{\text{f1}}^0 + b\Delta_{\text{f1}}^1 \\ &\quad + a\Delta_{\text{f2}}^0 + af_{01}\Delta_{\text{f2}}^1 + \Delta_{\text{left0}}^0 + \Delta_{\text{left2}}^0. \end{aligned}$$

In every branch used in the proof, the displayed weights are nonnegative, all fourteen coefficients of the relevant C_i are nonpositive, and the strict right side is positive. These assertions follow by direct expansion from the branch inequalities; the implementation performs the same expansion without simplification.

Scheme 1 is Lemma 5.1. In the complement $b > 0, f_{11} = 0, af_{01} < b$, scheme 2 is equivalent to

$$f_{10} = 0, \quad z_{11} \geq 1, \quad ad_{01} + bd_{11} \geq a(f_{01} + t_{01}).$$

Scheme 5 is equivalent there to

$$\begin{aligned} f_{00} &= 1, & f_{01} &= 0, & t_{11} &= 0, & t_{10} &\leq f_{10}, \\ 2a(1 - o_{00}) + bf_{10}(2 - o_{00}) - bo_{10} &\leq 0, \\ a(1 - z_{00}) + b(2f_{10} - d_{10} - z_{10}) &\leq 0, \\ t_{00} &\leq d_{00}, & t_{01} &\leq d_{01}. \end{aligned}$$

Schemes 3 and 6 are the special projection-form branches identified in Proposition 5.9; scheme 12 is the upper-form branch in Proposition 5.10.

Scheme 4 contains one positive integer parameter x . Every combined coefficient has the form $A_j + xB_j$. The implementation intersects the exact lower and upper integer bounds

$$x \geq \max_{B_j < 0} \left\lceil \frac{A_j}{-B_j} \right\rceil, \quad x \leq \min_{B_j > 0} \left\lfloor \frac{-A_j}{B_j} \right\rfloor,$$

together with $x \geq 1$, and then reconstructs the certificate. Thus no finite search limit is hidden in the parameterized family.

The machine-readable supports and all unsimplified coefficient predicates are distributed with the artifact as an additional check against transcription errors.

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